Electric Potential

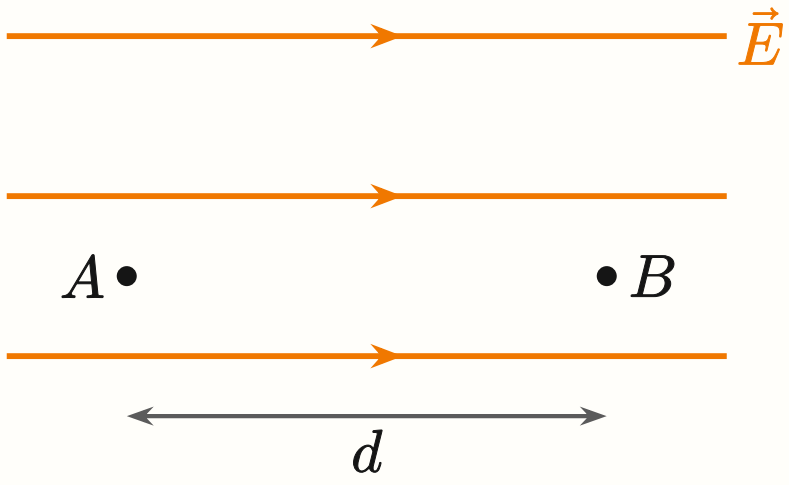
Class 12 · examples use $k = 9 \times 10^9\,\mathrm{N\,m^2C^{-2}}$, $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{C^2N^{-1}m^{-2}}$, $e = 1.6 \times 10^{-19}\,\mathrm{C}$

1 Definition

$$V = \frac{W_{\infty \to P}}{q_0} = \frac{U}{q_0}$$

W: work by an EXTERNAL agent moving a small test charge q_0 from infinity SLOWLY ($\Delta K = 0$) · SCALAR · $1\,\mathrm{V} = 1\,\mathrm{J\,C^{-1}}$, $[\mathrm{ML^2T^{-3}A^{-1}}]$

2 Potential difference



$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{l}$$

any path (conservative field)

$$V_A - V_B = Ed$$

UNIFORM $\vec{E}$, d along $\vec{E}$

Example

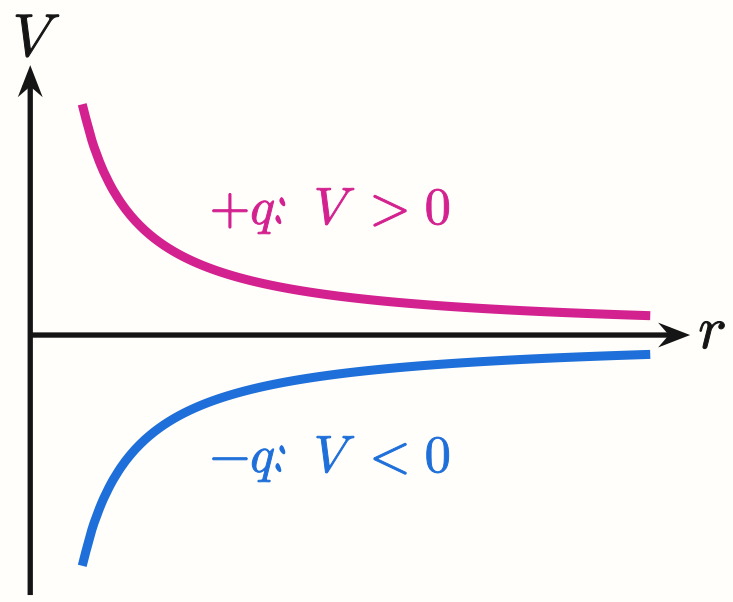
$E = 200\,\mathrm{V\,m^{-1}}$ *along* AB, $d = 0.5\,\mathrm{m}$: $V_A - V_B = 100\,\mathrm{V}$; *work by the FIELD on* $2\,\mu\mathrm{C}$ *from* A *to* B: $q(V_A - V_B) = 2 \times 10^{-4}\,\mathrm{J}$.

Golden points

★ *V FALLS along $\vec{E}$. From rest, electric force only: $+q$ moves to LOWER V, $-q$ to HIGHER V.*

Point Charge and Systems

1 Point charge



$$V = \frac{kq}{r} = \frac{q}{4\pi\varepsilon_0 r}$$

point charge, vacuum, $V(\infty) = 0$ · sign of q KEPT · medium everywhere: $\div\kappa$

Example

$q = 2\,\text{nC}$, $r = 0.3\,\text{m}$: $V = \dfrac{9\times10^{9}\times2\times10^{-9}}{0.3} = 60\,\text{V}$.

2 System of charges

$$V = k\sum_i \frac{q_i}{r_i}$$

ALGEBRAIC sum (scalar) · r_i: distance from q_i

Example

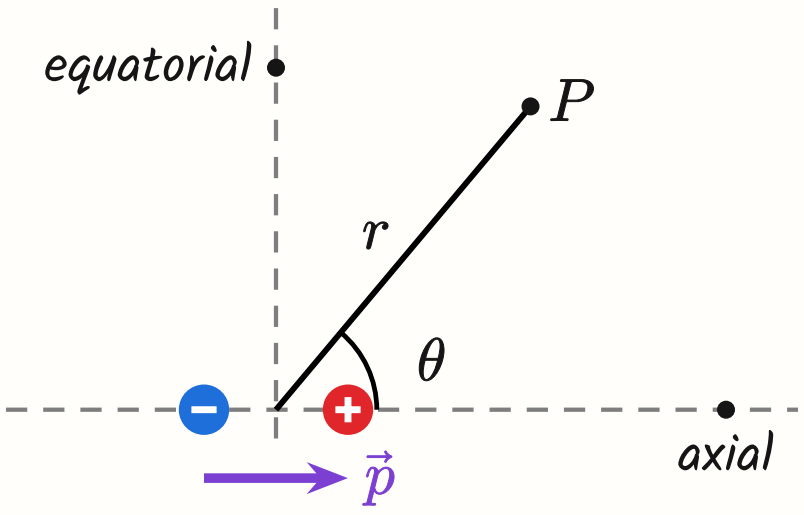
$+4\,\mu\text{C}$ *at* $x = 0$, $-2\,\mu\text{C}$ *at* $x = 6\,\text{m}$: $V = 0$ *on the axis where* $4/|x| = 2/|x-6|$: $x = 4\,\text{m}$ *and* $x = 12\,\text{m}$ *(beyond the SMALLER charge).*

Golden points

★ $V = 0 \not\Rightarrow E = 0$ ($+q, +q, -q, -q$ *round a square, like charges adjacent: centre* $V = 0$, $E \neq 0$). $E = 0 \not\Rightarrow V = 0$ *(midway between two equal* $+q$*).*

Potential of a Dipole

1 Any point



$$V = \frac{kp\cos\theta}{r^2}$$

$p = q(2a)$, from $-q$ to $+q$ · θ from $\vec{p}$ ·

SHORT dipole: $r \gg a$ · $V = \frac{k\vec{p}\cdot\vec{r}}{r^3}$

2 Axial and equatorial

$$V_{axial} = \frac{kp}{r^2 - a^2} \approx \frac{kp}{r^2}, \qquad V_{eq} = 0$$

axial: on the $+q$ side (on the $-q$ side V is negative) · r from the centre ·

$V_{eq} = 0$ EXACTLY at every distance

Example

$p = 1 \times 10^{-8}\,\mathrm{C\,m}$, $r = 0.3\,\mathrm{m}$ (short dipole): axial

$V = \frac{9 \times 10^9 \times 10^{-8}}{0.09} = 1000\,\mathrm{V}$; at $\theta = 60°$: $500\,\mathrm{V}$; equatorial: 0.

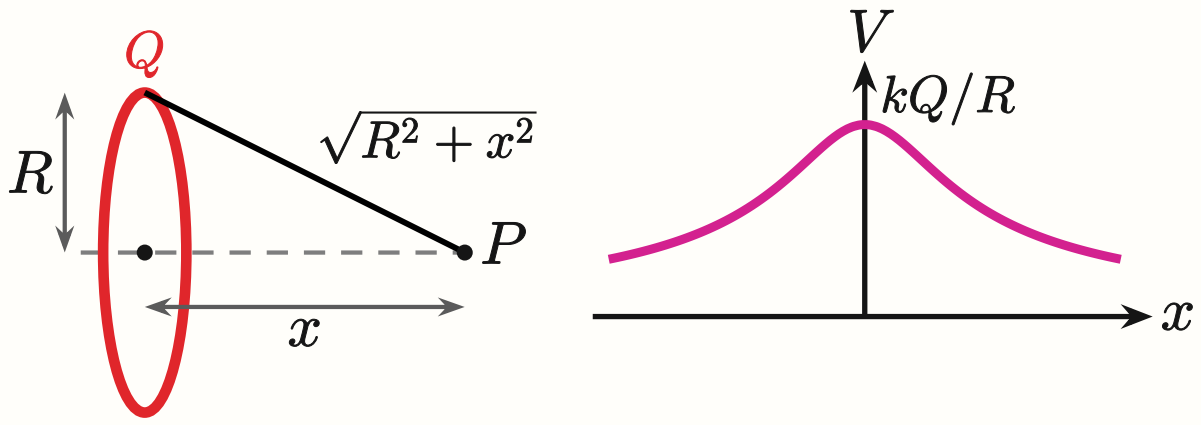
Golden points

★ Dipole: $V \propto 1/r^2$, $E \propto 1/r^3$ (point charge: $1/r$ and $1/r^2$).

★ Equatorial line: $V = 0$ but $E = kp/r^3$ (opposite $\vec{p}$, short dipole); moving a charge along it needs NO work.

Ring, Disc, Line and Sheet

1 Ring on its axis



$$V = \frac{kQ}{\sqrt{R^2+x^2}}, \qquad V_{centre} = \frac{kQ}{R}$$

all dq at the SAME distance: valid even for NON-uniform charge · any arc of charge Q: kQ/R at its centre

Example

$Q = 4\,\text{nC}$ $(kQ = 36\,\text{V m})$, $R = 0.3\,\text{m}$, $x = 0.4\,\text{m}$:
$V = 36/0.5 = 72\,\text{V}$; *centre* $36/0.3 = 120\,\text{V}$.

2 Disc on its axis

$$V = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{R^2+x^2} - x\right)$$

uniform σ, $x \geq 0$ · centre: $\sigma R/2\varepsilon_0$ · $x \gg R$: kQ/x

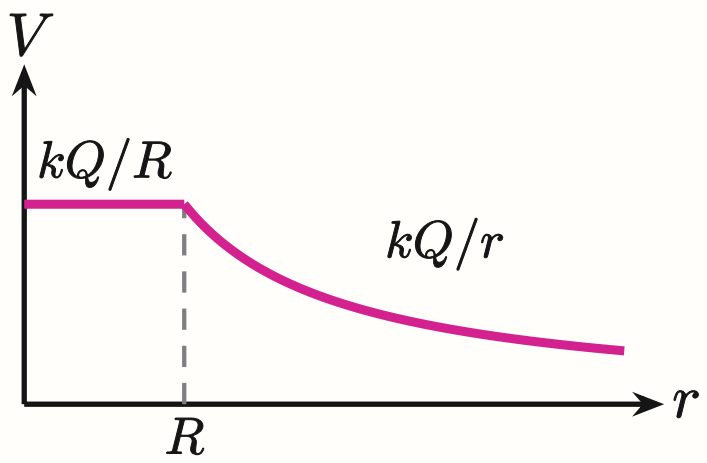
3 Infinite line and infinite sheet

$$V_1 - V_2 = 2k\lambda \ln\frac{r_2}{r_1}, \qquad V_1 - V_2 = \frac{\sigma}{2\varepsilon_0}(d_2 - d_1)$$

from $E = 2k\lambda/r$ and $\sigma/2\varepsilon_0$ · $V(\infty) = 0$ impossible here

Shells and Spheres

1 Spherical shell / conducting sphere

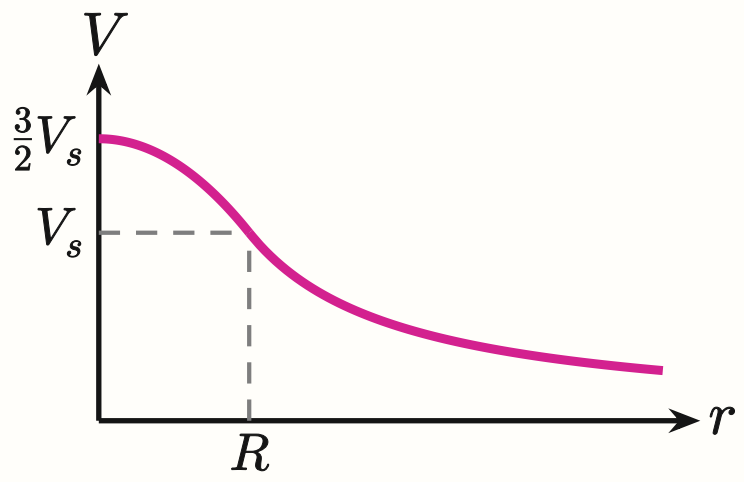


$$V = \frac{kQ}{R} \quad (r \le R)$$

$$V = \frac{kQ}{r} \quad (r \ge R)$$

thin shell with UNIFORM σ (conducting or not), or any conducting sphere (charge on its surface) · inside: V constant, $E = 0$

2 Uniform solid (non-conducting) sphere



$$V_{in} = \frac{kQ(3R^2 - r^2)}{2R^3}$$

uniform ρ, $r \le R$ · outside kQ/r · centre $\frac{3}{2}kQ/R$ = 1.5× surface value

Example

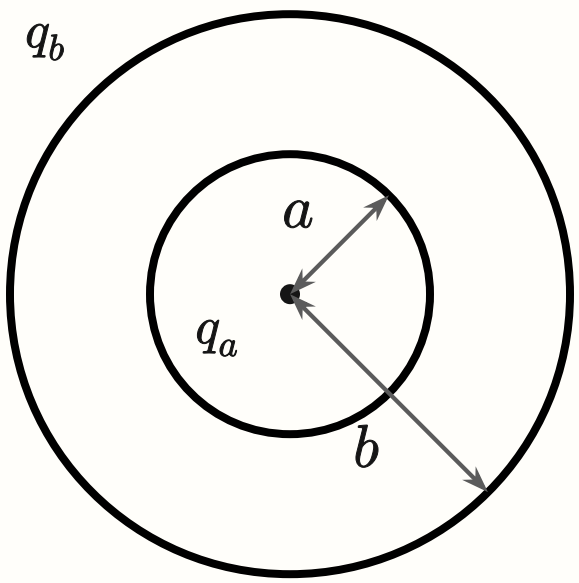
$Q = 1\,\text{nC}$, $R = 0.1\,\text{m}$: *surface* $90\,\text{V}$; *shell centre* $90\,\text{V}$; *solid-sphere centre* $135\,\text{V}$; *at* $r = 0.3\,\text{m}$ *(both)* $30\,\text{V}$.

Golden points

★ *n identical charged drops merge into one: $Q = nq$, $R = n^{1/3}r$, so $V = n^{2/3}V_0$ (27 drops at 10 V: 90 V); σ becomes $n^{1/3}$ times.*

Concentric Shells

1 Potential everywhere



$$r \le a: \quad V = k\left(\frac{q_a}{a} + \frac{q_b}{b}\right)$$

$$a \le r \le b: \quad V = k\left(\frac{q_a}{r} + \frac{q_b}{b}\right)$$

$$r \ge b: \quad V = \frac{k(q_a + q_b)}{r}$$

thin shells, charges uniform

$$V_a - V_b = kq_a\left(\frac{1}{a} - \frac{1}{b}\right)$$

independent of q_b: the OUTER charge never changes the difference

Example

$q_a = 2\,\text{nC}$, $a = 0.1\,\text{m}$; $q_b = 4\,\text{nC}$, $b = 0.2\,\text{m}$:

$V_a = 180 + 180 = 360\,\text{V}$, $V_b = \dfrac{9 \times 10^9 \times 6 \times 10^{-9}}{0.2} = 270\,\text{V}$,

$V_a - V_b = 90\,\text{V}$.

2 Earthing a shell

- *Earthed: $V = 0$, the CHARGE adjusts. Inner shell earthed, outer has Q: $\dfrac{kq}{a} + \dfrac{kQ}{b} = 0$, $q = -\dfrac{Qa}{b}$. With $a = 0.1\,\text{m}$, $b = 0.2\,\text{m}$, $Q = 6\,\text{nC}$: $q = -3\,\text{nC}$.*
- *Outer shell earthed: total charge inside it gets cancelled outside: its charge becomes $-q_a$ (no charges outside it).*

Field from Potential

1 Gradient

$$E_x = -\frac{\partial V}{\partial x},\quad E_y = -\frac{\partial V}{\partial y},\quad E_z = -\frac{\partial V}{\partial z}$$

$$\vec{E} = -\nabla V, \qquad E = -\frac{dV}{dr}$$

$\vec{E}$ points towards DECREASING V, along the steepest fall · dr along $\vec{E}$ · $1\,\mathrm{V\,m^{-1}} = 1\,\mathrm{N\,C^{-1}}$

Example

$V = (2x^2y - 3z)\,\mathrm{V}$ *(x, y, z in m):* $\vec{E} = -(4xy\,\hat{i} + 2x^2\,\hat{j} - 3\hat{k})$. *At* $(1, 2, 0)\,\mathrm{m}$: $\vec{E} = (-8\hat{i} - 2\hat{j} + 3\hat{k})\,\mathrm{V\,m^{-1}}$, $|\vec{E}| = \sqrt{77} \approx 8.77\,\mathrm{V\,m^{-1}}$.

2 Potential from field

$$V(P) = -\int_{\infty}^{P} \vec{E}\cdot d\vec{r}, \qquad V_A - V_B = \int_{A}^{B} \vec{E}\cdot d\vec{r}$$

electrostatic field (conservative) · any convenient path

Example

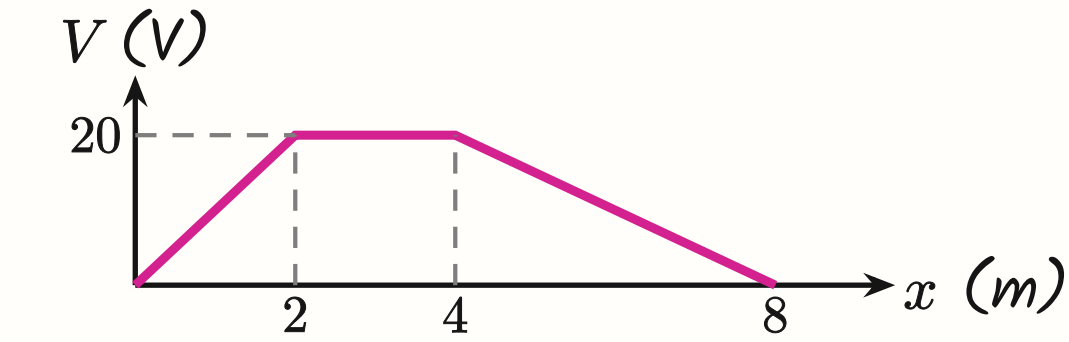
$\vec{E} = 10x\,\hat{i}\,\mathrm{V\,m^{-1}}$ *(x in m):* $V(0) - V(2\,\mathrm{m}) = \int_0^2 10x\,dx = 20\,\mathrm{V}$ *(potential falls along $+x$).*

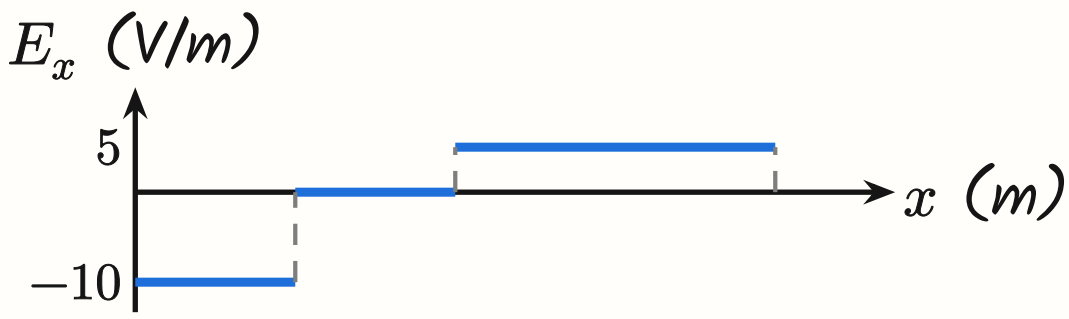
Golden points

★ *V constant in a region:* $\vec{E} = 0$ *there (inside a conductor).* $\vec{E} = 0$ *at a point: V stationary there, NOT necessarily zero. Uniform field:* $E = V/d$.

Reading V-x and E-x Graphs

1 From V to E: minus the slope





- Slope $+10\,\mathrm{V\,m^{-1}}$ (0–2 m): $E_x = -10\,\mathrm{V\,m^{-1}}$ (along $-x$); flat (2–4 m): $E_x = 0$; slope -5 (4–8 m): $E_x = +5\,\mathrm{V\,m^{-1}}$.

2 From E to V: minus the area

$$V_2 - V_1 = -\int_{x_1}^{x_2} E_x\,dx$$

minus the SIGNED area under the E_x-x graph · path along x

Example

Same graph, 4 to 8 m: area $5 \times 4 = 20\,\mathrm{V}$, so $V(8) - V(4) = -20\,\mathrm{V}$ ($20 \to 0\,\mathrm{V}$ as drawn).

Golden points

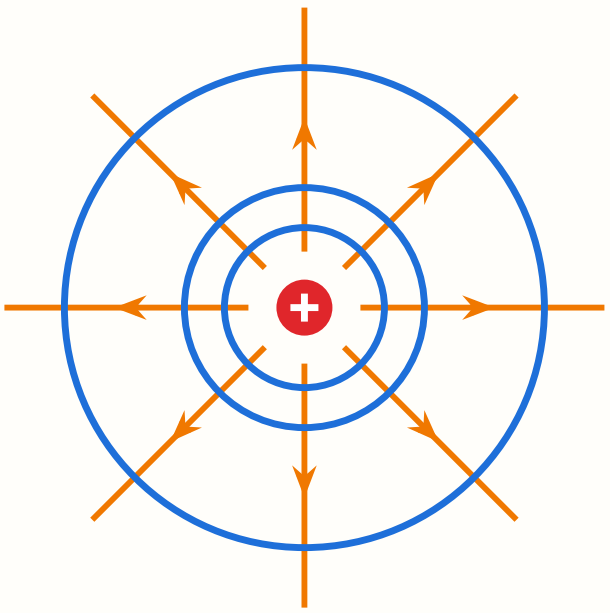
★ At a maximum or minimum of $V(x)$: $E_x = 0$. Steepest V-x slope: strongest field.

Equipotential Surfaces

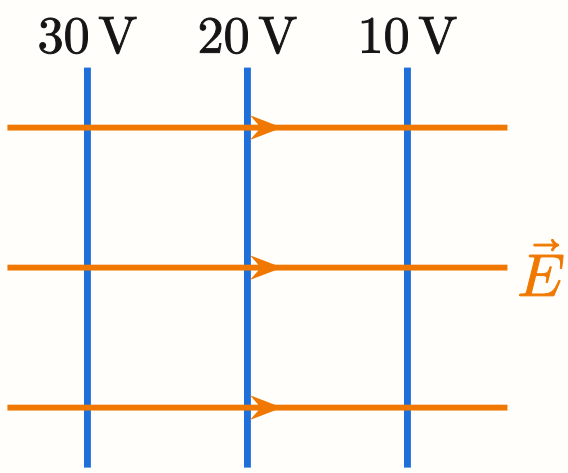
1 Properties

- Same V everywhere: NO work moving a charge along it.
- $\vec{E} \perp$ surface, from higher to lower V. Two never intersect.
- Equal steps ΔV: surfaces CLOSER where the field is stronger, $E \approx \Delta V / \Delta r$.

2 Shapes



point charge: spheres
r, $1.5r$, $3r$ for $3V_0$, $2V_0$, V_0



uniform field:
parallel planes

Example

Equipotential planes 30, 20, 10 V, 5 cm apart:

$$E = \frac{10}{0.05} = 200\,\mathrm{V\,m^{-1}},$$ from the 30 V plane towards the 10 V plane.

Golden points

★ Dipole: the equatorial plane is the $V = 0$ equipotential. A conductor's surface (equilibrium) is an equipotential.

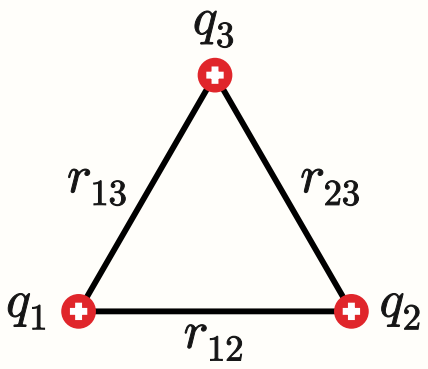
Potential Energy of Charges

1 Two charges

$$U = \frac{kq_1q_2}{r}$$

external work to bring them SLOWLY from infinity · like $U > 0$, unlike $U < 0$

2 Many charges



$$U = k\left(\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}}\right)$$

one term per PAIR (n charges: $n(n-1)/2$ pairs)

Example

Three $1\,\mu\text{C}$ charges, equilateral triangle of side $0.1\,\text{m}$:
$U = 3 \times \dfrac{9 \times 10^9 \times 10^{-12}}{0.1} = 0.27\,\text{J}$. *Four equal q, square of side a:*
$U \approx 5.41\,kq^2/a$.

3 Closest approach

$$\tfrac{1}{2}mv^2 = \frac{kqQ}{r_0}$$

head-on, target FIXED, like charges · r_0: closest approach

Example

$5\,\text{MeV}$ ($8 \times 10^{-13}\,\text{J}$) alpha ($2e$) at gold ($79e$):
$r_0 = k(2e)(79e)/K \approx 4.55 \times 10^{-14}\,\text{m}$.

Energy in an External Field

1 Charges in an external field

$$U = qV(\vec{r}), \qquad U = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{k q_1 q_2}{r_{12}}$$

V of EXTERNAL sources · field's work $A \to B$: $q(V_A - V_B)$

2 Dipole in a uniform field

$$U = -\vec{p} \cdot \vec{E} = -pE\cos\theta$$

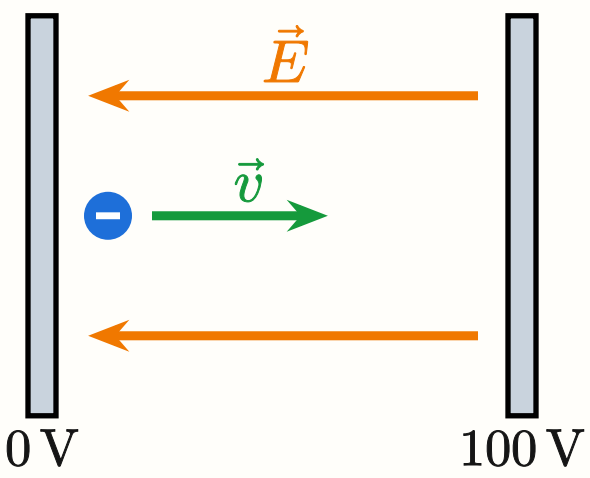
$$W_{ext} = pE(\cos\theta_1 - \cos\theta_2)$$

UNIFORM $\vec{E}$ · $U = 0$ at $90°$ · turned slowly $\theta_1 \to \theta_2$

Example

$p = 2 \times 10^{-8}\,\mathrm{C\,m}$, $E = 5 \times 10^{4}\,\mathrm{V\,m^{-1}}$: $0° \to 180°$ *needs* $2pE = 2 \times 10^{-3}\,\mathrm{J}$.

3 Accelerating a charge; electron-volt



$$K = q\,\Delta V$$

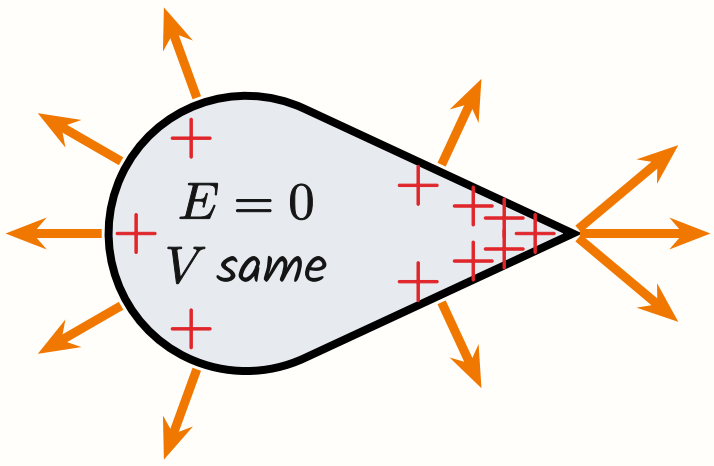
from rest, only the electric force, $v \ll c$ · $1\,\mathrm{eV} = 1.6 \times 10^{-19}\,\mathrm{J}$

Example

Electron through $100\,\mathrm{V}$: $K = 100\,\mathrm{eV} = 1.6 \times 10^{-17}\,\mathrm{J}$, $v \approx 5.93 \times 10^{6}\,\mathrm{m\,s^{-1}}$ $(m_e = 9.1 \times 10^{-31}\,\mathrm{kg})$.

Conductors in Electrostatics

1 Conductor in equilibrium



- $E = 0$ INSIDE the material.
- Whole conductor is ONE equipotential.
- Net charge sits on the OUTER surface.
- $\vec{E}$ just outside is $\perp$ the surface.
- Isolated conductor: σ larger where curvature is larger (sharp points).

$$E_{out} = \frac{\sigma}{\varepsilon_0}, \qquad \text{pressure} = \frac{\sigma^2}{2\varepsilon_0}$$

just outside the surface, along the outward normal · pressure always outward

2 Electrostatic shielding

- Cavity with NO charge inside a conductor: $E = 0$ in the cavity whatever the outside field (car or metal cage in lightning).
- Charge q placed in the cavity: $-q$ appears on the cavity wall, $+q$ on the outer surface (link: previous chapter).

Golden points

★ Sharp points have huge σ and E: air ionises (corona discharge): lightning conductors use pointed rods.

Earthing and Charge Sharing

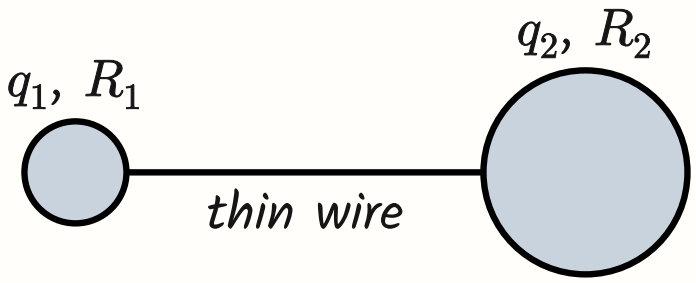
1 Earthing

- *Earth: huge conductor, taken as $V = 0$. An earthed conductor has $V = 0$; its charge is whatever makes $V = 0$ (NOT necessarily zero).*

Example

Earthed sphere, $R = 0.1\,\mathrm{m}$, point charge $6\,\mathrm{nC}$ at $0.3\,\mathrm{m}$ from its centre: $V_{centre} = \dfrac{kq}{d} + \dfrac{kq'}{R} = 0$, *induced* $q' = -\dfrac{qR}{d} = -2\,\mathrm{nC}$.

2 Two spheres joined by a wire



$$\frac{q_1}{q_2} = \frac{R_1}{R_2}, \qquad \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

far apart, wire's own charge neglected · COMMON potential

Example

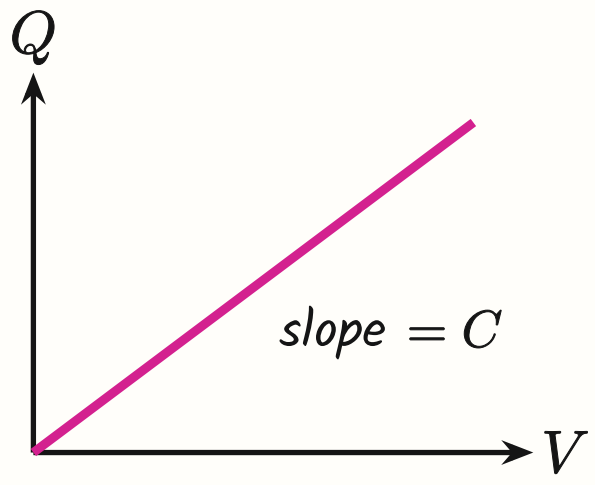
$R_1 = 2\,\mathrm{cm}$ *with* $6\,\mathrm{nC}$, $R_2 = 4\,\mathrm{cm}$ *uncharged:* $q_1 = 2\,\mathrm{nC}$, $q_2 = 4\,\mathrm{nC}$, *common* $V = 900\,\mathrm{V}$ *(was* $2700\,\mathrm{V}$*),* $\sigma_1/\sigma_2 = 2$. *Energy* $8.1 \to 2.7\,\mu\mathrm{J}$: $5.4\,\mu\mathrm{J}$ *lost as heat.*

Golden points

★ *Charge flows until the potentials are EQUAL, not the charges. The smaller sphere keeps less charge but a larger σ and surface field.*

Capacitance

1 Definition



$$C = \frac{Q}{V}$$

Q: charge on EITHER plate (magnitude) · V: potential difference · farad: $1\,\mathrm{F} = 1\,\mathrm{C\,V^{-1}}$, $[\mathrm{M^{-1}L^{-2}T^{4}A^{2}}]$

- C depends ONLY on size, shape, separation and the medium, not on Q or V.
- Capacitor: two conductors with $+Q$ and $-Q$. Symbol: two parallel plates.

2 Isolated sphere

$$C = 4\pi\varepsilon_0 R = \frac{R}{k}$$

other plate at infinity · in a medium filling all space: $\times\kappa$

Example

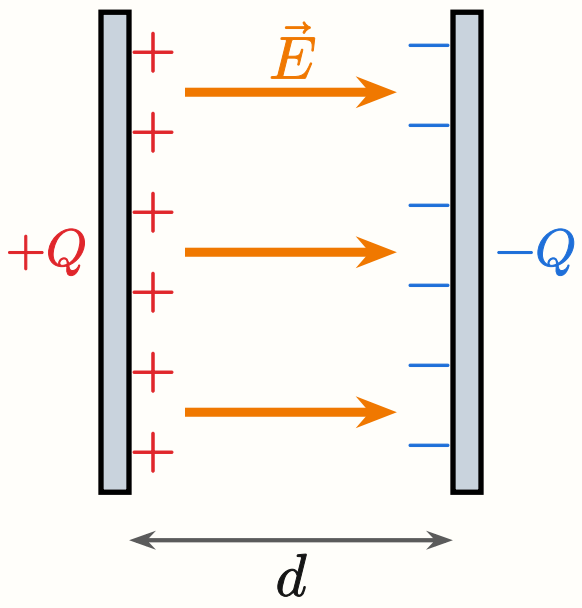
$R = 9\,\mathrm{cm}$: $C = \dfrac{0.09}{9\times10^{9}} = 10\,\mathrm{pF}$; at $100\,\mathrm{V}$, $Q = 1\,\mathrm{nC}$. Earth, $R = 6.4\times10^{6}\,\mathrm{m}$: $C \approx 711\,\mu\mathrm{F}$. A $1\,\mathrm{F}$ sphere needs $R = 9\times10^{9}\,\mathrm{m}$!

Golden points

★ Bringing an EARTHED conductor near a charged one lowers its potential for the same Q, so C rises: the idea of a capacitor.

Parallel Plate Capacitor

1 Field and capacitance



$$E = \frac{\sigma}{\varepsilon_0}, \quad V = Ed, \quad C = \frac{\varepsilon_0 A}{d}$$

$\sigma = Q/A$ · vacuum (air) between · $d \ll$ plate size (fringing neglected) · $E = 0$ outside

- Each plate alone gives $\sigma/2\varepsilon_0$: the two ADD between the plates and CANCEL outside.

Example

$A = 0.01\,\mathrm{m}^2$, $d = 1\,\mathrm{mm}$: $C = \dfrac{8.85 \times 10^{-12} \times 0.01}{10^{-3}} = 88.5\,\mathrm{pF}$. At $100\,\mathrm{V}$: $Q = 8.85\,\mathrm{nC}$, $E = 10^5\,\mathrm{V\,m^{-1}}$.

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2 n plates, alternate ones joined

$$C = \frac{(n-1)\varepsilon_0 A}{d}$$

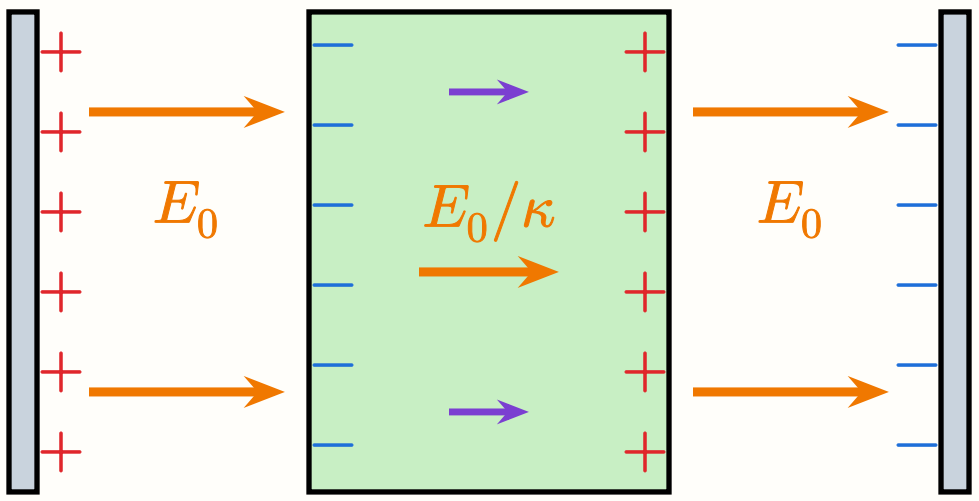
n identical plates, equal gaps d, alternate plates connected: $n-1$ capacitors in parallel

Example

5 plates with the area and gap above: $C = 4 \times 88.5 = 354\,\mathrm{pF}$.

Dielectrics and Polarisation

1 What a dielectric does



dielectric κ · purple: induced $\vec{p}$

- Non-polar molecules get INDUCED dipoles; polar ones ALIGN. Both give polarisation $\vec{P}$ (dipole moment per volume) along $\vec{E}$.
- Bound charges (− facing the + plate) oppose the field: inside, $E = E_0/\kappa$ (battery disconnected).

2 Key relations

$$\kappa = \frac{\varepsilon}{\varepsilon_0} = 1 + \chi, \quad \sigma_b = \sigma\left(1 - \frac{1}{\kappa}\right), \quad C = \kappa C_0$$

linear dielectric, $\vec{P} = \varepsilon_0 \chi \vec{E}$ · $C = \kappa C_0$ when it FILLS the gap · conductor: $\kappa \to \infty$

Example

$\kappa = 5$ filling the gap: C becomes $5C_0$; bound charge $\sigma_b = 0.8\,\sigma$.

Golden points

★ Dielectric strength: largest safe field (air $\approx 3 \times 10^6\ \mathrm{V\,m^{-1}}$); a good dielectric raises C AND the safe voltage.

Partly Filled Capacitors

1 Slab parallel to the plates

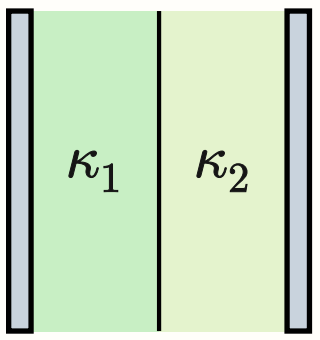
$$C = \frac{\varepsilon_0 A}{d - t + \dfrac{t}{\kappa}}, \qquad C_{metal} = \frac{\varepsilon_0 A}{d - t}$$

slab thickness t, full area · ANY position · metal: $\kappa \to \infty$

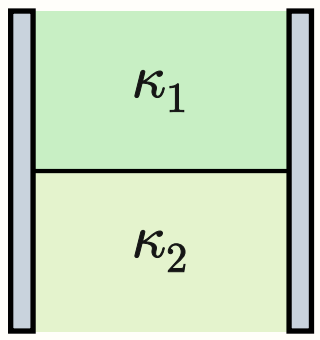
Example

$d = 4\,\text{mm}$, *slab* $t = 2\,\text{mm}$, $\kappa = 4$: $C = \dfrac{\varepsilon_0 A}{2.5\,\text{mm}} = 1.6\,C_0$. *Metal slab of* $2\,\text{mm}$: $2\,C_0$.

2 Layers and side-by-side



layers: SERIES



side by side: PARALLEL

$$C_{layers} = \frac{\varepsilon_0 A}{\dfrac{t_1}{\kappa_1} + \dfrac{t_2}{\kappa_2}}, \qquad C_{side} = \frac{\varepsilon_0 (\kappa_1 A_1 + \kappa_2 A_2)}{d}$$

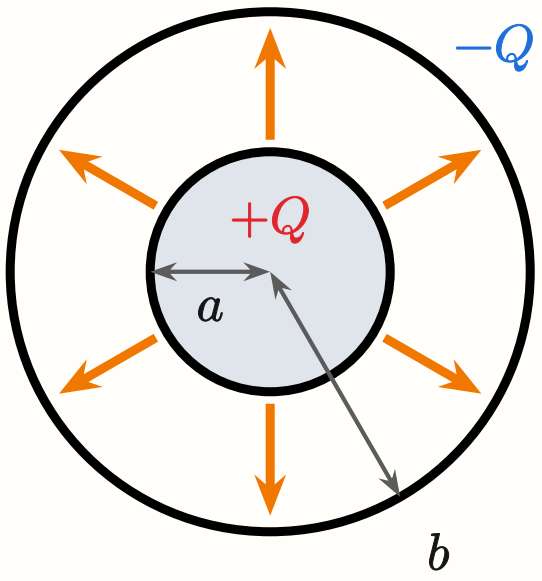
layers fill the area, thicknesses $t_1 + t_2 = d$ · side by side: each fills the gap d

Example

Halves with $\kappa = 4$ *and air side by side:* $2.5\,C_0$. *Two layers* $d/2$ *each,* $\kappa_1 = 2$, $\kappa_2 = 4$: $C = \frac{8}{3} C_0 \approx 2.67\,C_0$.

Spherical and Cylindrical

1 Spherical capacitor



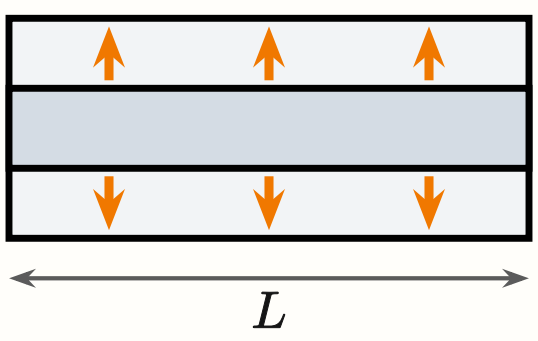
$$C = 4\pi\varepsilon_0 \frac{ab}{b-a}$$

inner radius a ($+Q$), outer b ($-Q$) · medium κ between: $\times\kappa$ · $b \to \infty$: $4\pi\varepsilon_0 a$

Example

$a = 9\,\text{cm}$, $b = 10\,\text{cm}$: $C = \dfrac{0.09 \times 0.1}{9 \times 10^9 \times 0.01} = 100\,\text{pF}$. *Inner EARTHED, charge given to the outer:* $C = 4\pi\varepsilon_0 \dfrac{b^2}{b-a} \approx 111\,\text{pF}$.

2 Cylindrical capacitor



$$C = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}$$

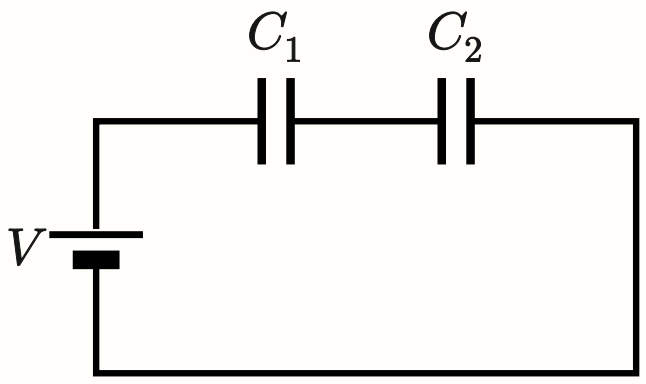
coaxial, radii $a < b$ · $L \gg b$ (end effects neglected) · medium κ: $\times\kappa$

Example

$L = 1\,\text{m}$, $b/a = e$: $C = 2\pi\varepsilon_0 \approx 55.6\,\text{pF}$.

Series and Parallel

1 Series



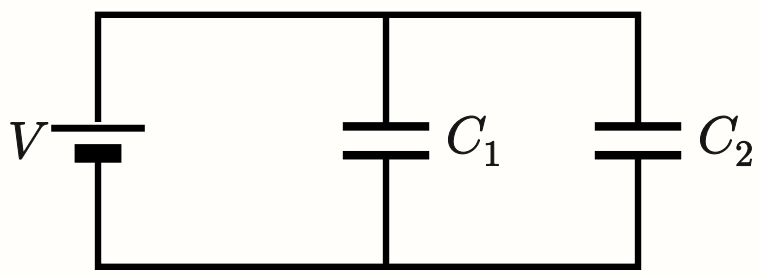
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

initially uncharged: SAME charge Q on each · V divides as $1/C$ · C below the smallest

Example

$2\,\mu\text{F}$ *and* $3\,\mu\text{F}$ *in series across* $10\,\text{V}$: $C = 1.2\,\mu\text{F}$, $Q = 12\,\mu\text{C}$, $V_1 = 6\,\text{V}$, $V_2 = 4\,\text{V}$.

2 Parallel



$$C = C_1 + C_2 + \dots$$

SAME V on each · Q divides as C

Example

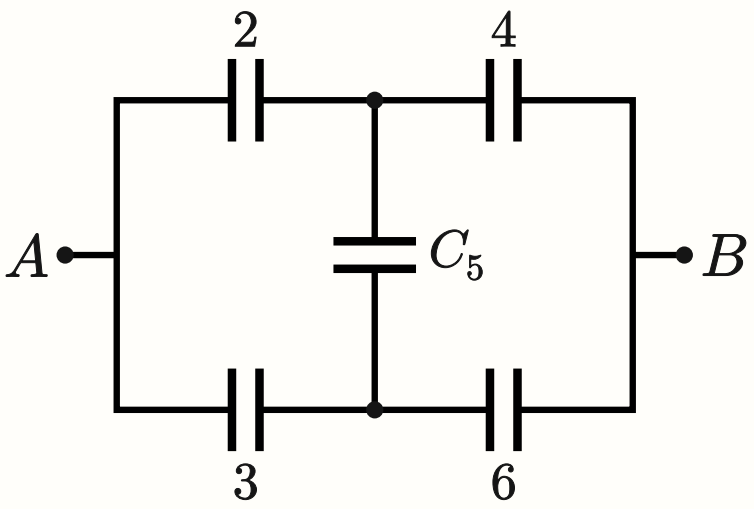
The same pair in parallel on $10\,\text{V}$: $C = 5\,\mu\text{F}$, $Q_1 = 20\,\mu\text{C}$, $Q_2 = 30\,\mu\text{C}$.

Golden points

★ *n identical capacitors C: series C/n, parallel nC. Capacitors combine OPPOSITE to resistors.*

Capacitor Networks

1 Balanced bridge



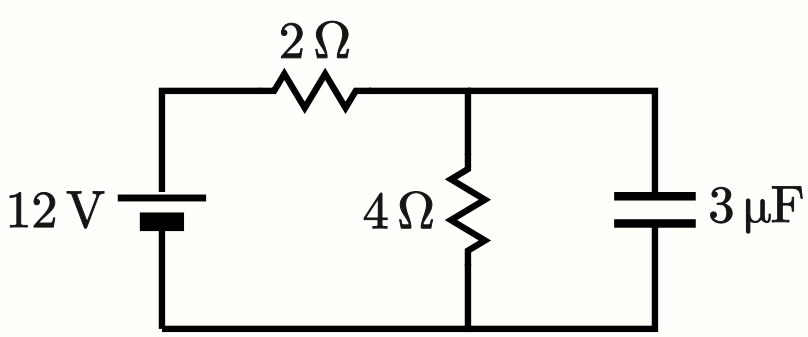
- $\dfrac{C_1}{C_2} = \dfrac{C_3}{C_4}$: *the middle capacitor gets NO charge; remove it.*
- *Here (μF):* $\dfrac{2\times 4}{6} + \dfrac{3\times 6}{9} = \frac{4}{3} + 2 \approx 3.33\,\mu\text{F}.$

2 Isolated node

$$\sum_{\text{node}} q = \textit{initial total}$$

node not connected to a battery terminal · zero if all started UNCHARGED · with loop rule $\sum \Delta V = 0$

3 Capacitor in a DC circuit



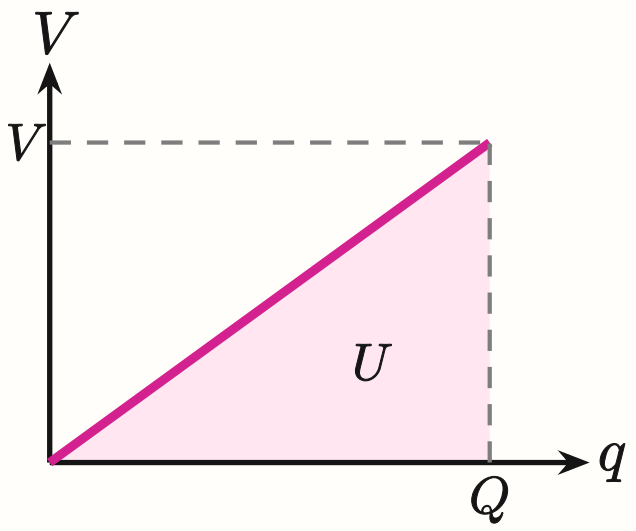
- *STEADY state: no current in the capacitor branch.*
- $I = 12/6 = 2\,\text{A}$; $V_C = V_{4\Omega} = 8\,\text{V}$; $Q = 24\,\mu\text{C}$.

Golden points

★ *Steady state: replace each capacitor by an OPEN gap, find the potentials, then* $Q = CV$ *for each.*

Energy Stored

1 Work to charge



$$U = \frac{Q^2}{2C} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV$$

area under V-q · $dW = V\,dq$ with $V = q/C$ · stored in the FIELD

Example

$4\,\mu\text{F}$ *at* $50\,\text{V}$: $Q = 200\,\mu\text{C}$, $U = \tfrac{1}{2} \times 4 \times 10^{-6} \times 50^2 = 5\,\text{mJ}$.

2 Energy density

$$u = \tfrac{1}{2}\varepsilon_0 E^2, \qquad u = \tfrac{1}{2}\kappa\varepsilon_0 E^2$$

energy per unit volume of ANY electric field (vacuum; dielectric κ) · plates: $U = u \times Ad$

Example

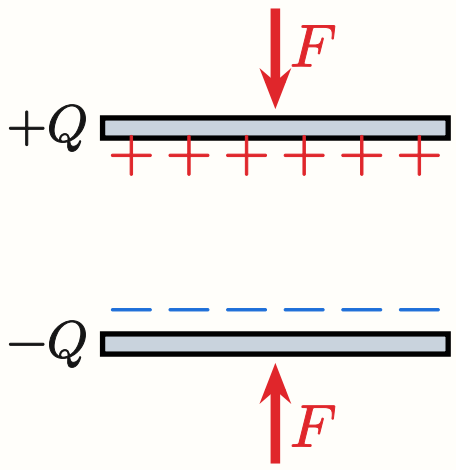
Air at its breakdown field $3 \times 10^6\,\text{V m}^{-1}$:
$u = \tfrac{1}{2} \times 8.85 \times 10^{-12} \times 9 \times 10^{12} \approx 39.8\,\text{J m}^{-3}$.

Golden points

★ *Series (initially uncharged): same Q, so $U = Q^2/2C$: SMALLER C stores more. Parallel: same V, so $U = \tfrac{1}{2}CV^2$: LARGER C stores more.*

Force Between the Plates

1 Attraction



$$F = \frac{Q^2}{2\varepsilon_0 A} = \tfrac{1}{2} Q E$$

$E = \sigma/\varepsilon_0$: field BETWEEN the plates · a plate feels only the OTHER plate's $\sigma/2\varepsilon_0$, hence the $\frac{1}{2}$

$$F = \frac{\varepsilon_0 A V^2}{2d^2}, \qquad \frac{F}{A} = \frac{\sigma^2}{2\varepsilon_0}$$

same force in terms of V · from $F = -dU/dx$ at constant Q

Example

$Q = 2\,\mu\text{C},\ A = 0.01\,\text{m}^2$: $F = \dfrac{(2\times10^{-6})^2}{2\times8.85\times10^{-12}\times0.01} \approx 22.6\,\text{N}.$

2 Force on a dielectric slab

$$F = \frac{\varepsilon_0 b(\kappa-1)V^2}{2d}$$

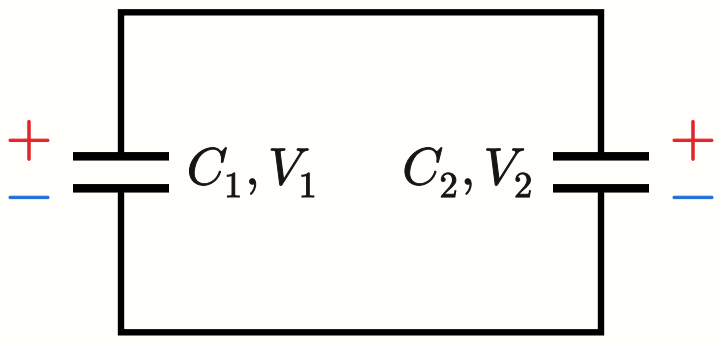
battery CONNECTED (V fixed) · plate width b, gap d · pulls the slab IN, whatever its depth

Golden points

★ *Battery REMOVED (constant Q): the pull on the slab falls as it goes in, $F = \dfrac{Q^2}{2C^2}\dfrac{dC}{dx}$.*

Joining Charged Capacitors

1 Common potential



$$V = \frac{C_1 V_1 \pm C_2 V_2}{C_1 + C_2}$$

charge conserved · +: LIKE plates joined, −: OPPOSITE plates joined (take magnitudes)

2 Energy lost as heat

$$\Delta U = \frac{C_1 C_2 \,(V_1 \mp V_2)^2}{2(C_1 + C_2)}$$

like plates: $(V_1 - V_2)^2$; opposite: $(V_1 + V_2)^2$ · SAME whatever the wire resistance (heat and radiation)

Example

$2\,\mu\text{F}$ *at* $100\,\text{V}$ *joined to an uncharged* $2\,\mu\text{F}$: $V = 50\,\text{V}$, U: $10 \to 5\,\text{mJ}$, *HALF lost.*

Example

$3\,\mu\text{F}$ *at* $100\,\text{V}$ *and* $6\,\mu\text{F}$ *at* $40\,\text{V}$. *Like plates:*

$$V = \frac{300 + 240}{9} = 60\,\text{V},$$ *heat* $3.6\,\text{mJ}$. *Opposite:*

$$V = \frac{300 - 240}{9} \approx 6.67\,\text{V},$$ *heat* $19.6\,\text{mJ}$.

Golden points

★ *Identical capacitors, one charged and one empty: always HALF the energy is lost, even as* $R \to 0$.

Battery Connected or Removed

1 Dielectric κ fills the gap

BATTERY	Q	V	C	E	U
Connected: V fixed	κQ	V	κC	E	κU
Removed: Q fixed	Q	V/κ	κC	E/κ	U/κ

2 Separation $d \to nd$

BATTERY	Q	V	C	E	U
Connected: V fixed	Q/n	V	C/n	E/n	U/n
Removed: Q fixed	Q	nV	C/n	E	nU

3 Where the energy goes

- V fixed: battery work $W_b = V\Delta Q = 2\,\Delta U$. Dielectric in: battery gives $2\Delta U$, field stores ΔU, rest goes into pulling the slab.
- Q fixed: no battery; external work $= \Delta U$ (pulling plates apart stores energy; the slab is sucked in, U falls).

Golden points

★ First ask: battery connected or not? connected $\Rightarrow$ V constant; removed (isolated) $\Rightarrow$ Q constant.

Charging Energy and Van de Graaff

1 Battery charges a capacitor

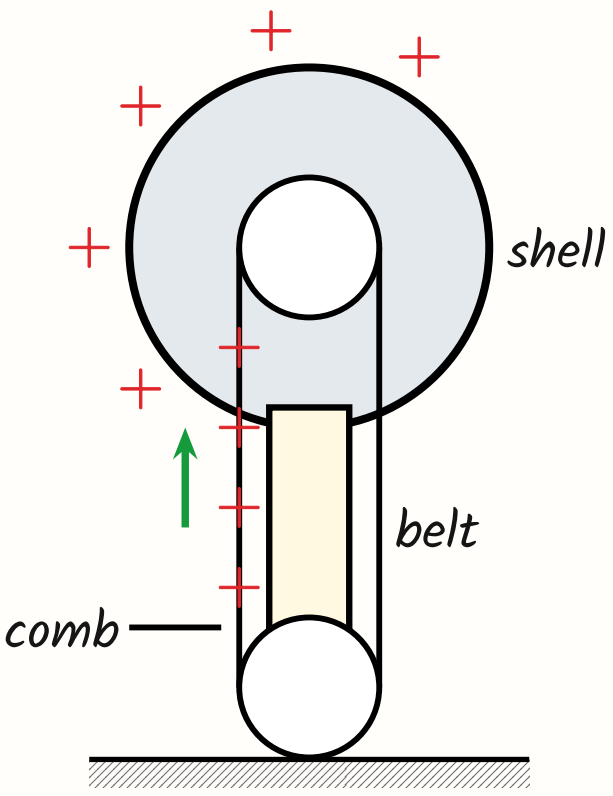
$$W_b = QV = CV^2, \quad U = \tfrac{1}{2}CV^2, \quad H = \tfrac{1}{2}CV^2$$

uncharged capacitor charged FULLY by a battery V through any resistance · half the battery's work is heat

Example

$10\,\mu\text{F}$ *by a* $20\,\text{V}$ *battery:* $W_b = 4\,\text{mJ}$, $U = 2\,\text{mJ}$, *heat* $2\,\text{mJ}$ *(for any* R*).*

2 Van de Graaff generator



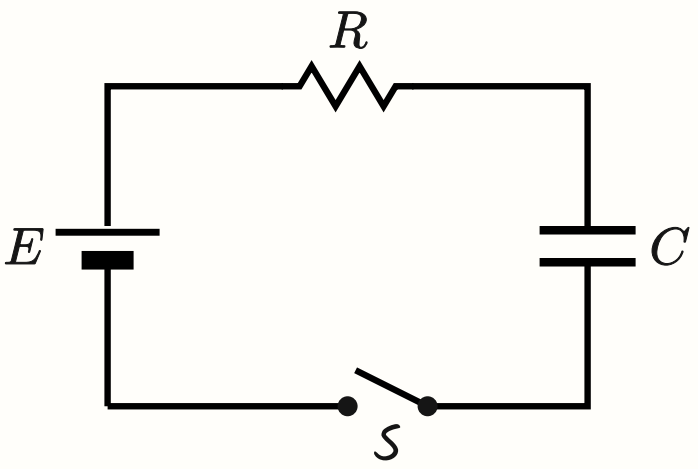
- *Charge on a small conductor INSIDE a shell always flows to the shell:* $V_{in} - V_{out} = kq\left(\frac{1}{r} - \frac{1}{R}\right) > 0$ *for* $q > 0$, *whatever the shell's charge.*
- *Belt carries charge up (sprayed on by a comb); the shell's potential rises to millions of volts.*
- *Limit: breakdown of air,* $V_{max} \approx E_b R$ ($R = 1\,\text{m}$: $3 \times 10^6\,\text{V}$). *Used to accelerate charged particles.*

Golden points

★ *Same principle as shielding: charge always goes to the OUTER surface of a conductor in equilibrium.*

RC Circuit: Charging

1 Charging through R



$$q = CE\left(1 - e^{-t/\tau}\right)$$

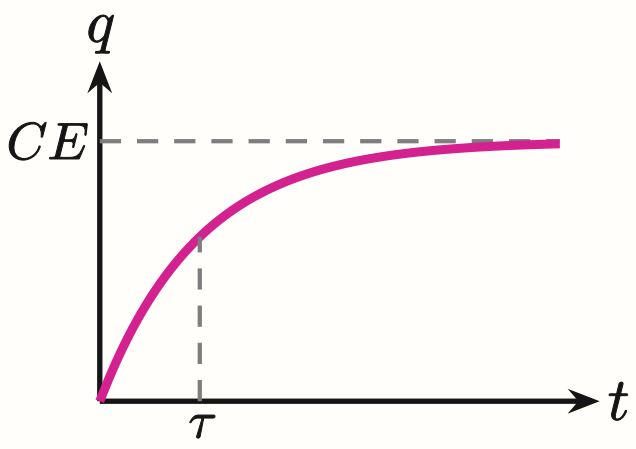
$$i = \frac{E}{R}e^{-t/\tau}$$

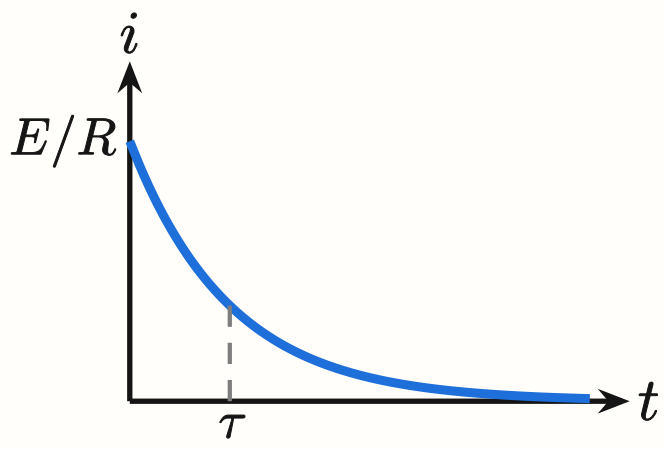
S closed at $t = 0$, C UNCHARGED at first

$$\tau = RC$$

$q = 63.2\%$ of CE at $t = \tau$

2 Graphs





Example

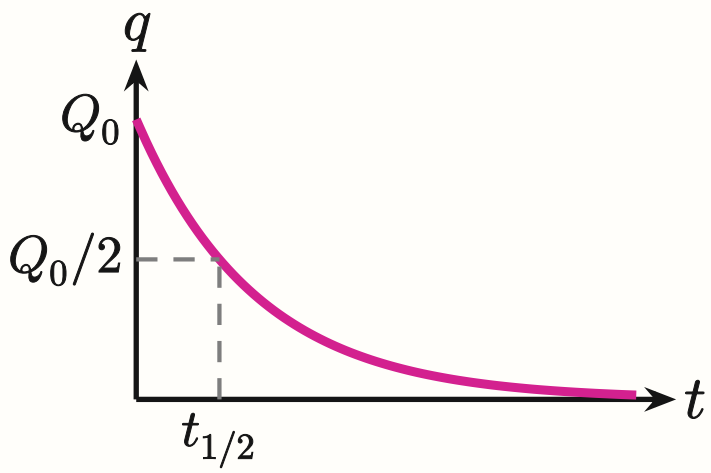
$R = 10\,\text{k}\Omega$, $C = 100\,\mu\text{F}$, $E = 10\,\text{V}$: $\tau = 1\,\text{s}$, *final* $Q = 1\,\text{mC}$, $i_0 = 1\,\text{mA}$; *at* $t = 1\,\text{s}$: $q \approx 0.632\,\text{mC}$, $i \approx 0.368\,\text{mA}$.

Golden points

★ *Battery supplies CE^2; capacitor stores $\frac{1}{2}CE^2$; the resistor heats by $\frac{1}{2}CE^2$ (any R).*

RC Circuit: Discharging

1 Discharging through R



$$q = Q_0 e^{-t/\tau}$$

$$|i| = \frac{Q_0}{RC} e^{-t/\tau}$$

charged capacitor joined to R at $t = 0$, no battery · $\tau = RC$ · $V_C = V_0 e^{-t/\tau}$

$$t_{1/2} = \tau \ln 2 \approx 0.693\,\tau, \qquad U = U_0 e^{-2t/\tau}$$

charge (and V, i) halve every $t_{1/2}$ · energy halves in $\frac{1}{2}\tau \ln 2 \approx 0.347\,\tau$

Example

$\tau = 2\,\mathrm{s}$: *charge falls to* 25% *at* $t = 2\tau \ln 2 \approx 2.77\,\mathrm{s}$; *energy to* 25% *at* $\tau \ln 2 \approx 1.39\,\mathrm{s}$.

2 Two instants to remember

- $t = 0$: *an UNCHARGED capacitor acts like a plain wire (short); its voltage cannot jump.*
- *Steady state* ($t \gg \tau$): *no current through it: an open switch, holding the voltage across its branch.*

Golden points

★ *Units:* $\Omega \times \mathrm{F} = \mathrm{s}$. *Larger* R *or* C: *slower charging AND discharging.*

Common Traps

1 Exam traps

Avoid these

- ✘ $V = 0$ means $E = 0$ (equatorial line of a dipole: $V = 0$, $E \neq 0$).
- ✘ Potential inside a charged conducting shell is zero (it equals the SURFACE value; only $E = 0$).
- ✘ Earthed means uncharged (earthed means $V = 0$).
- ✘ Plate force $= QE$ (it is $\frac{1}{2}QE$ with E between the plates).
- ✘ A dielectric always lowers E (only with the battery REMOVED; connected: $E = V/d$ unchanged).
- ✘ C grows with Q (C depends only on geometry and medium).
- ✘ Joining charged capacitors conserves energy (charge yes; energy is lost as heat unless they were at the same V).
- ✘ Series capacitors always carry equal charge (only if initially uncharged).
- ✘ An electron moves to lower V (released from rest, electric force only: to HIGHER V).

2 Solving recipe

Steps

- Potential: add kq/r as SCALARS; field: $\vec{E} = -\nabla V$.
- Capacitors: fix what is constant (Q or V), then C, then U.

Summary

$$V = \frac{kq}{r} \quad \bullet \quad \vec{E} = -\nabla V \quad \bullet \quad C = \frac{\varepsilon_0 A}{d} \quad \bullet \quad U = \frac{Q^2}{2C}$$

Questions: JEE Main Level (I)

Q1 · JEE Main level · Numerical

A ring of radius $0.3\,\mathrm{m}$ carries $5\,\mathrm{nC}$. Work (in $\mu\mathrm{J}$) to bring a $2\,\mu\mathrm{C}$ charge slowly from infinity to its centre?

Solution: $V_{centre} = kQ/R = 45/0.3 = 150\,\mathrm{V}$; $W = qV = 3\times10^{-4}\,\mathrm{J}$.

Answer: 300

Q2 · JEE Main level · MCQ

$V = (6xy - y + 2yz)\,\mathrm{V}$ (x, y, z in m). $|\vec{E}|$ at $(1,1,0)\,\mathrm{m}$: (A) $\sqrt{65}\,\mathrm{V\,m^{-1}}$ (B) $\sqrt{29}\,\mathrm{V\,m^{-1}}$ (C) $7\,\mathrm{V\,m^{-1}}$ (D) $\sqrt{77}\,\mathrm{V\,m^{-1}}$

Solution: $\vec{E} = -(6y\,\hat{i} + (6x - 1 + 2z)\,\hat{j} + 2y\,\hat{k}) = -(6\hat{i} + 5\hat{j} + 2\hat{k})$; $|\vec{E}| = \sqrt{36 + 25 + 4} = \sqrt{65}$.

Answer: (A)

Q3 · JEE Main level · MCQ

$3\,\mu\mathrm{F}$ and $6\,\mu\mathrm{F}$ (uncharged) in series across $9\,\mathrm{V}$. Charge on each and voltage on the $3\,\mu\mathrm{F}$: (A) $18\,\mu\mathrm{C}$, $6\,\mathrm{V}$ (B) $18\,\mu\mathrm{C}$, $3\,\mathrm{V}$ (C) $27\,\mu\mathrm{C}$, $9\,\mathrm{V}$ (D) $54\,\mu\mathrm{C}$, $6\,\mathrm{V}$

Solution: $C = \dfrac{3\times6}{9} = 2\,\mu\mathrm{F}$, $Q = 18\,\mu\mathrm{C}$, $V_3 = 18/3 = 6\,\mathrm{V}$.

Answer: (A)

Q4 · JEE Main level · Numerical

A $10\,\mu\mathrm{F}$ capacitor is charged to $100\,\mathrm{V}$; the battery is REMOVED and a slab of $\kappa = 2$ fills the gap. New stored energy in mJ?

Solution: $U_0 = \tfrac{1}{2}CV^2 = 50\,\mathrm{mJ}$; Q fixed, $C \to 2C$: $U = U_0/\kappa = 25\,\mathrm{mJ}$.

Answer: 25

Questions: JEE Main Level (2)

Q5 · JEE Main level · MCQ

A proton and an alpha particle start from rest and are accelerated through the same potential difference (non-relativistic). $v_p : v_\alpha$ is:

(A) $1:\sqrt{2}$ (B) $\sqrt{2}:1$ (C) $2:1$ (D) $1:2$

Solution: $\frac{1}{2}mv^2 = qV$, $v \propto \sqrt{q/m}$: $\sqrt{1/1} : \sqrt{2/4} = \sqrt{2}:1$.

Answer: (B)

Q6 · JEE Main level · Numerical

A $4\,\mu\text{F}$ capacitor charged to $50\,\text{V}$ is joined to an uncharged $6\,\mu\text{F}$ capacitor. Heat produced in mJ?

Solution: $V = 200/10 = 20\,\text{V}$; U: $5 \to 2\,\text{mJ}$, so heat $= 3\,\text{mJ}$.

Answer: 3

Q7 · JEE Main level · MCQ

27 identical charged mercury drops, each at $10\,\text{V}$, merge into one drop. Its potential: (A) $30\,\text{V}$ (B) $90\,\text{V}$ (C) $270\,\text{V}$ (D) $10\,\text{V}$

Solution: $Q = 27q$, $R = 3r$: $V = \dfrac{27}{3} \times 10 = 90\,\text{V}$ ($n^{2/3}V_0$).

Answer: (B)

Q8 · JEE Main level · Numerical

An air capacitor of $9\,\text{pF}$ has a $6\,\text{mm}$ gap. A slab $3\,\text{mm}$ thick, $\kappa = 4$, is placed in it parallel to the plates. New C in pF?

Solution: $C = C_0 \dfrac{d}{d - t + t/\kappa} = 9 \times \dfrac{6}{3 + 0.75} = 14.4\,\text{pF}$.

Answer: 14.4

Questions: JEE Advanced Level (I)

A1 · JEE Advanced level · One or more correct

Concentric thin conducting shells: radius R with charge Q, radius $2R$ with charge $2Q$. Correct: (A) inner shell at $2kQ/R$ (B) outer shell at $3kQ/2R$ (C) if the OUTER shell is earthed, the inner one is at $kQ/2R$ (D) if the INNER shell is earthed, its charge becomes $-2Q$

Solution: $V_{in} = \dfrac{kQ}{R} + \dfrac{k(2Q)}{2R} = \dfrac{2kQ}{R}$; $V_{out} = \dfrac{k(3Q)}{2R}$. Outer earthed: its charge becomes $-Q$, $V_{in} = \dfrac{kQ}{R} - \dfrac{kQ}{2R} = \dfrac{kQ}{2R}$. Inner earthed: $\dfrac{kq}{R} + \dfrac{k(2Q)}{2R} = 0$, $q = -Q$, not $-2Q$.

Answer: (A), (B), (C)

A2 · JEE Advanced level · Numerical

An uncharged $2\,\mu\text{F}$ capacitor is charged by a battery through $1\,\text{M}\Omega$. After how many seconds is its stored energy one quarter of the final value?

Solution: $\tau = RC = 2\,\text{s}$. $U \propto q^2$: $U = \frac{1}{4}U_f$ when $q = \frac{1}{2}Q$: $1 - e^{-t/\tau} = \frac{1}{2}$, $t = \tau \ln 2 \approx 1.39\,\text{s}$.

Answer: ≈ 1.39

Tip

★ Earthing a shell: write $V = 0$ for THAT shell with its charge unknown; the other charges stay fixed.

Questions: JEE Advanced Level (2)

battery and capacitor ideal

A3 · JEE Advanced level · Paragraph

A $5\,\mu\text{F}$ parallel plate capacitor stays CONNECTED to a $20\,\text{V}$ battery while its plate separation is slowly doubled. Find (i) the change in stored energy, (ii) the work done by the battery, (iii) the work done by the external agent.

Solution: $C \to 2.5\,\mu\text{F}$, Q: $100 \to 50\,\mu\text{C}$. (i) U: $1000 \to 500\,\mu\text{J}$, $\Delta U = -500\,\mu\text{J}$. (ii) $W_b = V\Delta Q = 20 \times (-50) = -1000\,\mu\text{J}$ (charge pushed back into the battery). (iii) $W_{ext} = \Delta U - W_b = +500\,\mu\text{J}$.

Answer: (i) $-500\,\mu\text{J}$ (ii) $-1000\,\mu\text{J}$ (iii) $+500\,\mu\text{J}$

A4 · JEE Advanced level · Matching

Parallel plate capacitor. (P) slab $\kappa = 3$ fills the gap, battery connected (Q) same slab, battery removed (R) separation doubled, battery connected (S) separation doubled, battery removed. Match with (1) charge increases (2) voltage decreases (3) energy halves (4) voltage doubles.

Solution: (P) $Q \to 3Q$. (Q) $V \to V/3$, $U \to U/3$. (R) $Q \to Q/2$, $U = \frac{1}{2}CV^2 \to U/2$. (S) $V \to 2V$, $U \to 2U$.

Answer: P–1, Q–2, R–3, S–4

Tip

★ Battery work is $V\Delta Q$ (negative when charge is pushed back); the agent supplies the rest: $W_{ext} = \Delta U - W_b$.

Questions: JEE Advanced Level (3)

AS · JEE Advanced level · One or more correct

A non-conducting sphere of radius R has charge Q spread uniformly through its volume. Correct: (A) $V_{centre} = 1.5\,V_{surface}$ (B) E is largest at the surface (C) $V(R/2) = 11kQ/8R$ (D) $E(R/2) = E(2R)$

Solution: $V_{in} = kQ(3R^2 - r^2)/2R^3$: centre $3kQ/2R$; at $R/2$: $kQ(3 - \frac{1}{4})/2R = 11kQ/8R$. $E = kQr/R^3$ inside (rises), kQ/r^2 outside (falls): max at R.

$E(R/2) = kQ/2R^2 \neq E(2R) = kQ/4R^2$.

Answer: (A), (B), (C)

A6 · JEE Advanced level · One or more correct

A $4\,\mu\text{F}$ capacitor charged to $50\,\text{V}$ discharges through $5\,\text{k}\Omega$. Correct: (A) initial current $10\,\text{mA}$ (B) charge halves after $\approx 13.9\,\text{ms}$ (C) energy halves after $\approx 6.93\,\text{ms}$ (D) total heat in the resistor is $10\,\text{mJ}$

Solution: $\tau = RC = 20\,\text{ms}$; $i_0 = 50/5000 = 10\,\text{mA}$; $t_{1/2} = \tau \ln 2 \approx 13.9\,\text{ms}$; $U \propto e^{-2t/\tau}$ halves at $\frac{1}{2}\tau \ln 2 \approx 6.93\,\text{ms}$; heat $= U_0 = \frac{1}{2}CV^2 = 5\,\text{mJ}$.

Answer: (A), (B), (C)

How to use these

- ★ Try each question before reading its solution; redo the ones you missed after a day.
- ★ Check every answer a second way (energy and charge, or a limiting case).